Risk-Aware Water-Surface Obstacle Detection and Warning via Segmentation and Monocular Depth Estimation

Aileen Yan
tayanwu@umich.edu

Kevin Zhou
yrzhou@umich.edu

Zheng Wang
zzww@umich.edu

I. Introduction

Autonomous Surface Vehicles (ASVs) are increasingly deployed in complex, high-traffic environments such as near-shore ports, marinas, and inland waterways. These operational domains present significant perception challenges due to highly dynamic water surfaces, strong reflections, sun glitter, waves, and the frequent presence of low-profile or partially submerged obstacles (e.g., floating debris, buoys, swimmers). Unlike structured urban roadways, the maritime environment lacks consistent ground planes or static reference features, complicating standard computer vision tasks. Failure to detect and respond to these hazards in a timely manner can result in severe safety incidents, including collisions and groundings.

To enable safe navigation, many existing ASV platforms rely on specialized sensing modalities such as marine radar, LiDAR, or sonar to detect obstacles and estimate their distances. While these sensors provide valuable geometric information, they significantly increase system cost, power consumption, and integration complexity, making them impractical for small, low-cost, or resource-constrained surface vehicles. In contrast, monocular cameras offer a lightweight and inexpensive sensing alternative.

Recent research in vision-based maritime perception has made significant progress in semantic segmentation, predominantly focusing on improving pixel-level accuracy metrics such as mean Intersection-over-Union (mIoU). However, in safety-critical scenarios, high mIoU does not inherently translate into navigational safety. The central question is not merely whether an obstacle is perfectly segmented at the pixel level, but whether it poses an immediate collision risk that should trigger a warning or evasive action. False alarms can degrade operator trust and system usability, while missed detections of nearby hazards can have catastrophic consequences.

Motivated by these challenges, this project proposes a risk-aware water-surface obstacle detection and warning framework via semantic segmentation and monocular depth estimation. Rather than relying on specialized ranging sensors during deployment, our approach fuses spatial semantic information with depth cues inferred from a single camera. To overcome the terrestrial-to-maritime domain gap inherent in off-the-shelf depth models, we calibrate monocular depth to a metric scale using radar ground truth, enabling physically grounded risk assessment.

By reasoning about obstacle proximity, lateral threat, and semantic class, the system generates discrete and interpretable warning signals (e.g., Safe, Caution, Immediate Warning) suitable for autonomous control modules or human operators. The framework is systematically evaluated on real-world maritime imagery, explicitly prioritizing safety-oriented metrics—such as Obstacle Recall and inference latency—over traditional segmentation accuracy alone.

II. Technical Approach

Our proposed system will follow a modular perception pipeline consisting of three main modules: *Semantic Segmentation*, *Monocular Depth Estimation*, and *Risk-Aware Sensor Fusion*.

A. *Semantic Segmentation*

1) Dataset: We will primarily utilize the LaRS dataset [1], a recently proposed and diverse benchmark for maritime obstacle perception. The dataset contains real-world surface navigation videos captured under challenging visual conditions, including strong water reflections, sun glitter, waves, and low-visibility obstacles such as partially submerged objects. To establish a rigorous baseline and directly address domain-specific challenges, we will supplement our training pipeline with the MaSTr1325 dataset [2]. While LaRS will serve as our primary evaluation benchmark for assessing the downstream effectiveness of risk-aware obstacle warning, incorporating MaSTr1325 will provide additional diverse coastal imagery to further enhance model robustness against complex water-surface phenomena.

2) Method: The semantic segmentation module serves as the foundational perception layer in our pipeline, responsible for delineating navigable water surfaces and isolating obstacle candidates at the pixel level. To build a robust system capable of handling complex maritime visual artifacts, such as reflections and wakes, we implemented a comparative evaluation framework using the LaRS dataset. Our approach systematically trains and benchmarks several representative architectures to identify the optimal backbone for our specific deployment constraints.

Our evaluation spans both general-purpose models and maritime-specific networks to analyze the trade-offs between

segmentation accuracy, computational efficiency, and architectural complexity:

- **SegFormer** [3]: A transformer-based segmentation model that leverages a lightweight and efficient design, allowing us to evaluate the efficacy of self-attention mechanisms on dynamic water surface reflections.
- **DeepLab** [4]: A robust CNN-based approach utilizing atrous spatial pyramid pooling (ASPP) to capture multi-scale contextual information, serving as our primary general-purpose convolutional baseline.
- **U-Net** [5]: A classic convolutional encoder–decoder architecture known for its strong baseline performance and architectural simplicity.
- **WaSR (Water Segmentation and Refinement)** [2]: A state-of-the-art, maritime-specific network explicitly designed to handle water wakes and sun glitter. This will be integrated to serve as our domain-specific upper bound for comparison.

To ensure a fair comparison, all candidate models are trained and validated under a unified pipeline. Our performance assessment extends beyond standard pixel-level metrics, such as mean Intersection-over-Union (mIoU), to prioritize safety-critical maritime metrics. Specifically, we evaluate Obstacle Recall to measure the system's ability to minimize missed hazards, and Obstacle Precision to quantify false alarm rates. Combined with inference latency measurements, these metrics will drive the selection of the final segmentation backbone, which will ultimately provide the spatial features required for the downstream monocular depth estimation and risk-aware warning modules.

3) Current Progress: At this stage, we have completed the initial semantic segmentation portion of the project. We prepared the LaRS dataset, verified image-mask alignment, and confirmed the valid semantic labels used in training. Since the raw images have varying resolutions, we implemented a unified preprocessing pipeline with fixed-size resizing for training and nearest-neighbor interpolation for segmentation masks. This pipeline allows us to consistently evaluate safety-oriented metrics, such as obstacle recall, across the entire benchmark.

Second, we implemented the end-to-end semantic segmentation training and evaluation pipeline. This includes the custom dataset loader, training loop, and a comprehensive metric computation module. In addition to standard pixel-level metrics like mIoU, we implemented safety-oriented metrics—specifically Obstacle Recall and Obstacle Precision—to quantify "missed hazards" and false alarm rates, respectively. Furthermore, we measured Inference Latency to ensure the architectures meet the real-time requirements for deployment on resource-constrained vehicles.

Third, we implemented two segmentation baselines, DeepLab and SegFormer, and trained both models under the same dataset and evaluation setup. This gives us an initial comparison between a CNN-based baseline and a transformer-based architecture. We further implemented visualization tools that save predicted masks and overlay them on the original RGB images for qualitative inspection. At this point, the segmentation component of the project is functional, and the current results provide a basis for selecting the backbone to be used in the next project stage. Detailed quantitative and qualitative comparisons are presented in the Results section.

B. Monocular Depth Estimation

1) Dataset: We use the sample subset of the WaterScenes dataset [6]. The full WaterScenes release provides rich multi-modal data and annotations, including camera images, radar, calibration, GPS, IMU, and various segmentation and detection labels. In this project, we focus on the following components:

- image: RGB images (`.jpg`) used as model input.
- radar/: radar point files (`.csv`) containing attributes such as Doppler, azimuth/elevation, 3D coordinates, labels, and instance IDs. We only use `u`, `v`, and `range`.

For each timestamp, radar points are projected onto image pixel coordinates (u, v), and the radar `range` is treated as ground-truth distance (GT) for evaluation. This setup enables a direct comparison between monocular depth predictions and radar-based distance measurements, while leaving other modalities for future multimodal fusion.

2) Method: We evaluate three variants of Depth Anything V2 (`small`, `base`, and `large`) [7]. The original Depth Anything models primarily predict relative depth. To obtain metric-scale predictions, we use the Depth-Anything-V2-Metric-Outdoor model, which is fine-tuned on synthetic Virtual KITTI data [8] for outdoor metric depth estimation.

However, Virtual KITTI is designed for autonomous driving scenarios (e.g., roads, vehicles, and urban layouts), which differ significantly from water-surface environments. As a result, applying this model directly to water scenes introduces a noticeable domain gap, leading to systematic estimation errors.

To mitigate this issue, we introduce a simple global scaling factor. Empirically, we observe that the predicted distances (`est`) exhibit a consistent scale mismatch with radar ground truth (`GT`). We therefore apply:

$$\texttt{est_scaled} = \texttt{est} \times \texttt{est_scale},$$

where `est_scale` is set to 0.4.

3) Future Work: Despite the introduction of the global scaling factor, the overall performance of the Depth Anything V2 metric model remains limited in this setting. This suggests that simple post-hoc calibration is insufficient to fully address the domain gap between training data and water-surface environments. Therefore, future work will focus on exploring more robust approaches to improve depth estimation accuracy, such as domain adaptation, range-aware calibration, or incorporating additional sensing modalities.

At the same time, we recognize the inherent limitations of monocular metric depth estimation, especially in challenging environments with limited texture and ambiguous geometry. As a result, rather than solely focusing on minimizing estimation error, the system will be designed to tolerate and account

for such uncertainties. This includes developing strategies that are robust to imperfect depth predictions, enabling safer and more reliable decision-making under uncertainty.

C. Risk-Aware Obstacle Warning

Outputs from semantic segmentation and monocular depth estimation will be fused to reason about obstacle risk. Drawing inspiration from established methodologies in autonomous driving [9], we will adopt a principled approach to risk assessment rather than relying on pure heuristics. We will adapt metrics such as Time-to-Collision (TTC) or worst-case proximity assumptions to the ASV domain. For each obstacle region, spatial location and relative depth information will be used to calculate a continuous risk score. To map this score to discrete warning levels (e.g., safe, caution, immediate warning) suitable for autonomous control modules or human operators, we will rigorously tune our warning thresholds using a dedicated validation split of the dataset. This systematic tuning process is critical to balancing the trade-off between false alarm rates and missed detection rates—our core safety-oriented evaluation metrics.

D. Dynamic Object Tracking and Kinematic Risk Assessment

While static obstacle detection is crucial, maritime environments are inherently dynamic. To address moving hazards (e.g., other vessels) and calculate accurate collision risks, we will extend our pipeline to process continuous video sequences. We will implement SORT (Simple Online and Realtime Tracking) [10], a highly efficient Multi-Object Tracking (MOT) framework, to associate segmented obstacle regions across frames and estimate their relative motion vectors. SORT's lightweight design introduces minimal computational overhead, making it ideal for the resource constraints of real-time ASV deployment.

Furthermore, realistic risk assessment depends heavily on the ASV's own kinematic state. We will factor in the ego-vehicle's speed to compute dynamic, maritime-specific safety metrics, most notably the Time to Closest Point of Approach (TCPA) [11] and Distance to Closest Point of Approach (DCPA). By transitioning from static distance heuristics to true kinematic risk evaluation, our warning generation will be robust to dynamic encounter scenarios without sacrificing the computational efficiency necessary for field operations.

E. Temporal Smoothing

Raw frame-by-frame predictions can be noisy due to perception uncertainty. Building upon the object tracks established in the previous module, we will implement temporal smoothing over consecutive frames. We will apply a moving average or a simple Kalman filter to both the estimated depth and the calculated risk scores of tracked objects. This prevents single-frame noise from triggering spurious warnings and ensures that warnings are issued and released in a stable, interpretable manner.

Fig. 1. Overview of the proposed perception pipeline. From top to bottom: input RGB image from the LaRS dataset, semantic segmentation highlighting water-surface regions and obstacle candidates, and monocular depth estimation providing relative depth cues for risk-aware obstacle warning.

F. Model Selection and Evaluation

Multiple segmentation and depth estimation models will initially be evaluated and compared. The final model combination will be selected based on a balance of perception accuracy, computational efficiency, deployment complexity, and inference latency.

Evaluation will emphasize safety-oriented metrics, including missed hazardous obstacles, warning delay, and false alarm rate, rather than pixel-level accuracy metrics alone.

III. Results

A. Semantic Segmentation

1) Quantitative Analysis: We evaluated two semantic segmentation baselines, DeepLab and SegFormer, on the LaRS validation split. To better reflect the safety-oriented goal of the project, the primary focus was on Obstacle Recall, which

serves as our proxy for "missed hazards," and Inference Latency, which determines real-time viability.

TABLE I
COMPARISON OF DEEPLAB AND SEGFORMER ON THE LARS VALIDATION SET.

	DeepLab	SegFormer
mIoU	0.8648	0.9659
Obstacle Recall	0.7983	0.9651
Obstacle Precision	0.9648	0.9737
Avg Latency (ms)	15.44	29.74

The results in Table I indicate that SegFormer provides superior safety performance, achieving a 96.51% Obstacle Recall. This is a significant improvement over the 79.83% Recall of the DeepLab baseline, meaning SegFormer is far less likely to miss hazardous objects like buoys or debris. While the SegFormer architecture exhibits a higher average latency of 29.74 ms compared to the DeepLab baseline, it remains highly viable for deployment on autonomous surface vehicles. For a boat moving at typical maritime speeds, a processing time of 29.74 ms (roughly 33 FPS) is still well within the limits for real-time safety and collision avoidance.

2) Qualitative Visualization: Figures 2 and 3 show a qualitative comparison of DeepLab and SegFormer on a validation image. Both models correctly segment the dominant sky and water regions, indicating that they capture the overall scene layout well. The main difference lies in the thin obstacle region near the waterline. DeepLab only recovers part of this structure and produces a more fragmented prediction, whereas SegFormer preserves a larger and more continuous obstacle region that more closely matches the ground truth. This observation is consistent with the quantitative results, especially the stronger obstacle recall achieved by SegFormer.

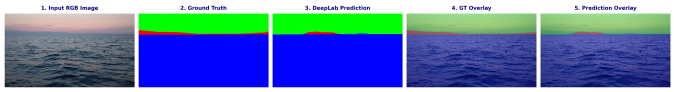

Fig. 2. Performance of DeepLab on a maritime scene.

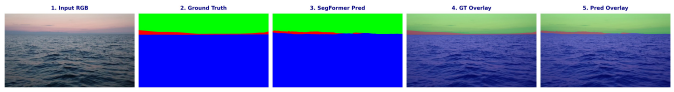

Fig. 3. Performance of SegFormer on a maritime scene.

B. Monocular Depth Estimation

We evaluate three Depth Anything model variants (`small`, `base`, `large`) on 33,648 radar-aligned points. The average inference time per image is 0.149 s, 0.279 s, and 0.710 s for the `small`, `base`, and `large` models, respectively. Tables II, III, and IV report performance across different distance ranges for each model.

To directly compare the three models in the near-range regime, Table V reports performance within 20 m.

TABLE II
DISTANCE-BINNED PERFORMANCE FOR THE SMALL MODEL.

Range (m)	Count	MAE (m)	Est/GT
0–10	2636	2.572	0.936
0–20	8041	4.434	0.787
0–50	19789	12.755	0.600
ALL	33648	39.210	0.441

TABLE III
DISTANCE-BINNED PERFORMANCE FOR THE BASE MODEL.

Range (m)	Count	MAE (m)	Est/GT
0–10	2636	2.298	1.118
0–20	8041	4.039	0.887
0–50	19789	11.783	0.666
ALL	33648	37.614	0.493

The results reveal several important trends. First, all models achieve substantially better performance at short ranges, while errors increase rapidly as distance grows, indicating that long-range depth estimation remains challenging in this setting. This behavior is also reflected in Fig. 4 and Fig. 5, where predictions increasingly deviate from the ideal diagonal as distance increases. Among the three variants, the `large` model consistently achieves the best accuracy across all ranges, particularly within 20 m, where it reaches near-metric scaling (Est/GT ≈ 1). In contrast, the `small` and `base` models tend to under-estimate depth even at moderate distances, suggesting limited capacity to recover accurate metric scale. However, this accuracy improvement comes at a significant computational cost, as the `large` model requires substantially longer inference time per image. The `base` model provides a reasonable trade-off between accuracy and efficiency, while the `small` model offers the fastest inference but the lowest accuracy. Overall, these results highlight a clear accuracy–efficiency trade-off, as well as the difficulty of generalizing monocular depth models to long-range water-surface environments.

Additionally, the Depth Anything V2 outdoor metric model imposes an upper bound on predicted depth at approximately 80 m. After applying the global scaling factor (0.4), this effectively limits the maximum estimated distance to around 32 m. This saturation effect is observable in both Fig. 4 and Fig. 5, where predictions flatten beyond this range. As a result, distances greater than 32 m are systematically clipped, leading to large errors in long-range regions. However, if the evaluation is restricted to near-range objects (within 20 m), this limitation has minimal impact, and the overall accuracy remains relatively high.

IV. REMAINING MILESTONES

Week 3–5: Pending Integration Tasks

- Set up the MaSTr1325 dataset for training diversity and integrate the WaSR model as the maritime-specific semantic segmentation baseline.

Week 6: Risk Modeling and Warning Generation

- Adapt risk assessment methodologies to the ASV domain.

TABLE IV
DISTANCE-BINNED PERFORMANCE FOR THE LARGE MODEL.

Range (m)	Count	MAE (m)	Est/GT
0–10	2636	2.365	1.260
0–20	8041	3.215	1.008
0–50	19789	9.907	0.766
ALL	33648	35.440	0.565

TABLE V
MODEL COMPARISON WITHIN 0–20 M.

Model	MAE (m)	RMSE (m)	Est/GT
Small	4.434	5.719	0.787
Base	4.039	5.363	0.887
Large	3.215	4.471	1.008

- Design the risk scoring function combining semantic weights, lateral risk, and the newly calibrated metric depth.
- Tune warning thresholds systematically on a dataset validation split. This will leverage the experimentally validated 20 m reliable depth range to optimize false alarm and missed detection rates.
- Implement discrete warning levels (Safe, Caution, Immediate Warning).

Week 7: Dynamic Tracking and Kinematic Integration

- Implement a lightweight multi-object tracking (MOT) module (e.g., SORT) to estimate obstacle motion vectors across continuous video frames.
- Integrate ego-vehicle speed into the risk scoring function to calculate true Time-to-Collision (TTC).
- Implement temporal smoothing over depth estimates and risk scores, and evaluate computational efficiency (e.g., measuring FPS overhead).
- Refine temporal parameters to reduce flickering and false alarms.

Week 8: Evaluation and Analysis

- Quantitatively evaluate the end-to-end system performance using core safety-oriented metrics.
- Compare the proposed depth-aware warning system against the segmentation-only baseline.
- Perform qualitative analysis on challenging maritime scenarios (e.g., sun glitter, wakes).

Week 9: Final Report and Presentation

- Compile experimental results, tables, and visualizations.
- Write the final report and summarize key findings regarding the accuracy-efficiency trade-offs identified during model selection.
- Prepare the final presentation and demonstration.

Fig. 4. Qualitative visualization of monocular depth estimation. From left to right: input RGB image with projected radar points, and the corresponding predicted depth map.

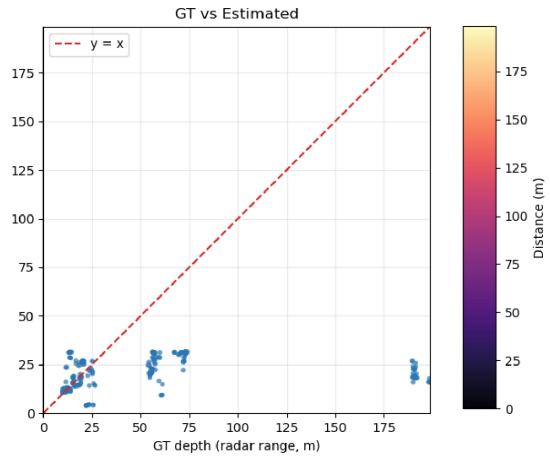

Fig. 5. Estimated depth versus radar ground truth. The deviation from the diagonal line indicates systematic under-estimation, especially at longer distances.

REFERENCES

[1] L. Žust, J. Perš, and M. Kristan, "Lars: A diverse panoptic maritime obstacle detection dataset and benchmark," 2023.

[2] B. Bovcon and M. Kristan, "Wasr—a water segmentation and refinement maritime obstacle detection network," *IEEE Transactions on Cybernetics*, vol. 52, no. 12, pp. 12661–12674, 2021.

[3] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "Segformer: Simple and efficient design for semantic segmentation with transformers," 2021.

[4] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017.

[5] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," 2015.

[6] S. Yao, R. Guan, Z. Wu, Y. Ni, Z. Huang, R. W. Liu, Y. Yue, W. Ding, E. G. Lim, H. Seo, K. L. Man, J. Ma, X. Zhu, and Y. Yue, "Waterscenes: A multi-task 4d radar-camera fusion dataset and benchmarks for autonomous driving on water surfaces," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 11, pp. 16584–16598, 2024.

[7] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao, "Depth anything v2," 2024.

[8] A. Gaidon, Q. Wang, Y. Cabon, and E. Vig, "Virtual worlds as proxy for multi-object tracking analysis," in *CVPR*, 2016.

[9] D. Lu, H. Du, Z. Wu, and S. Yang, "Risk assessment in autonomous driving: a comprehensive survey of risk sources, methodologies, and system architectures," *Autonomous Intelligent Systems*, vol. 5, no. 1, p. 24, 2025.

[10] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple online and realtime tracking," in *2016 IEEE international conference on image processing (ICIP)*, pp. 3464–3468, Ieee, 2016.

[11] Y. Qu and L. Cai, "Real-time emergency collision avoidance for unmanned surface vehicles with colregs flexibly obeyed," *Journal of Marine Science and Engineering*, vol. 10, no. 12, p. 2025, 2022.